plot the upper bound provided by the theorem, and in the third column, we use the upper bound on the distance to $\rho^*$ to give a lower or upper bound on the kernel sequence curve. One of the most important takeaways is that for all activation functions except exp, the distances $|\rho_\ell - \rho^*|$, shown on the log-scale, decay linearly with depth $\ell$. This is perfectly aligned with the prediction of Theorem 1, because exp is the only activation function that has polynomial convergence (See Table S.1). The gap between theory and empirical values corresponds to the worst-case analysis for the global convergence rate.

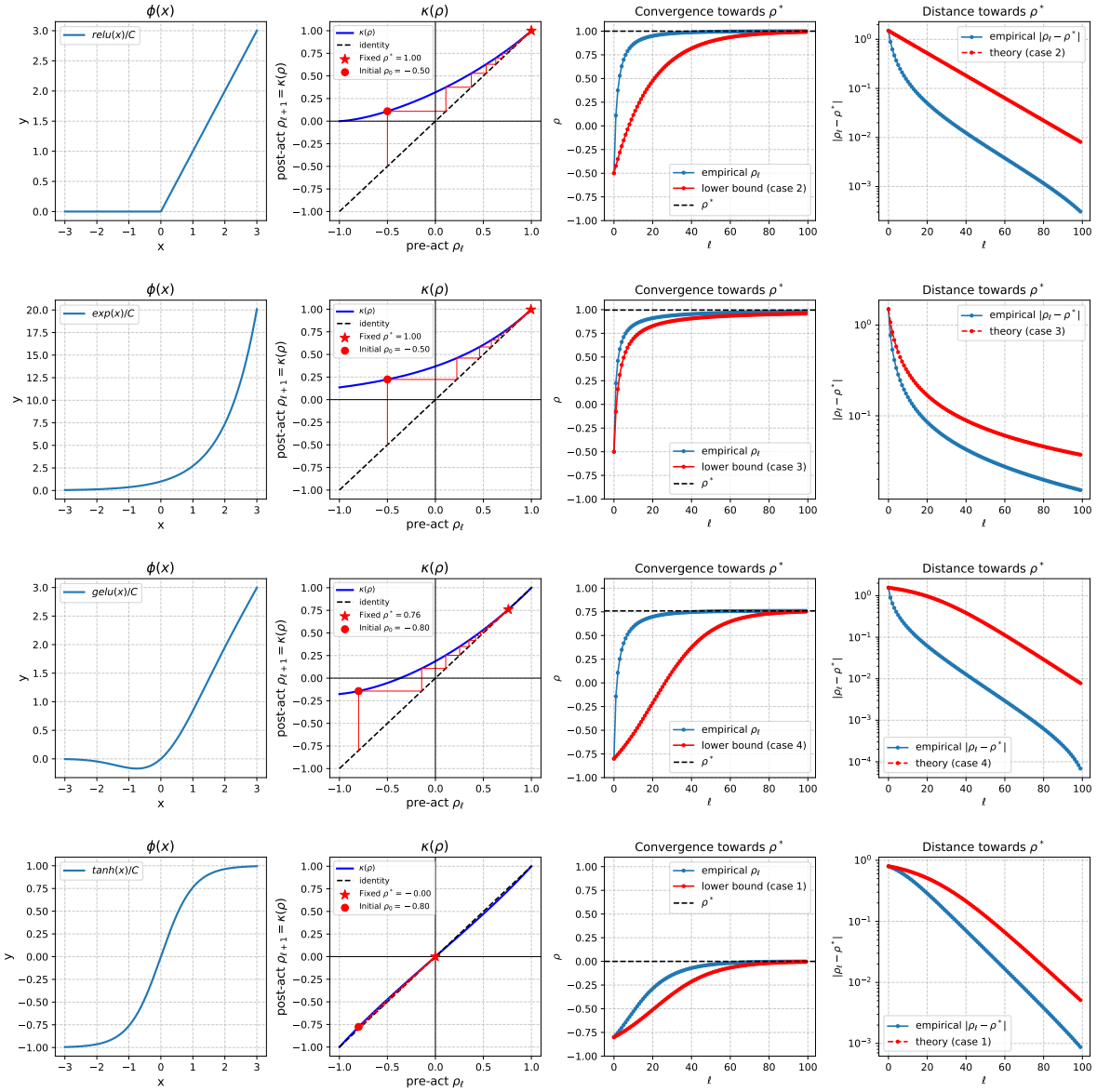


Figure S.1: Validation of Theorem 1 Each row corresponds to an activation, scaled down by a factor $C$ to obey $\mathbb{E}\,\phi(X)^2 = 1$.. From op to bottom: relu, exp, gelu, tanh. From left, the first column shows the activation, second column shows kernel map, third column shows the kernel sequence vs depth along with the theory prediction, and fourth column shows the distance to the fixed points in for theory and empirical kernels. (Remainder on the next page)

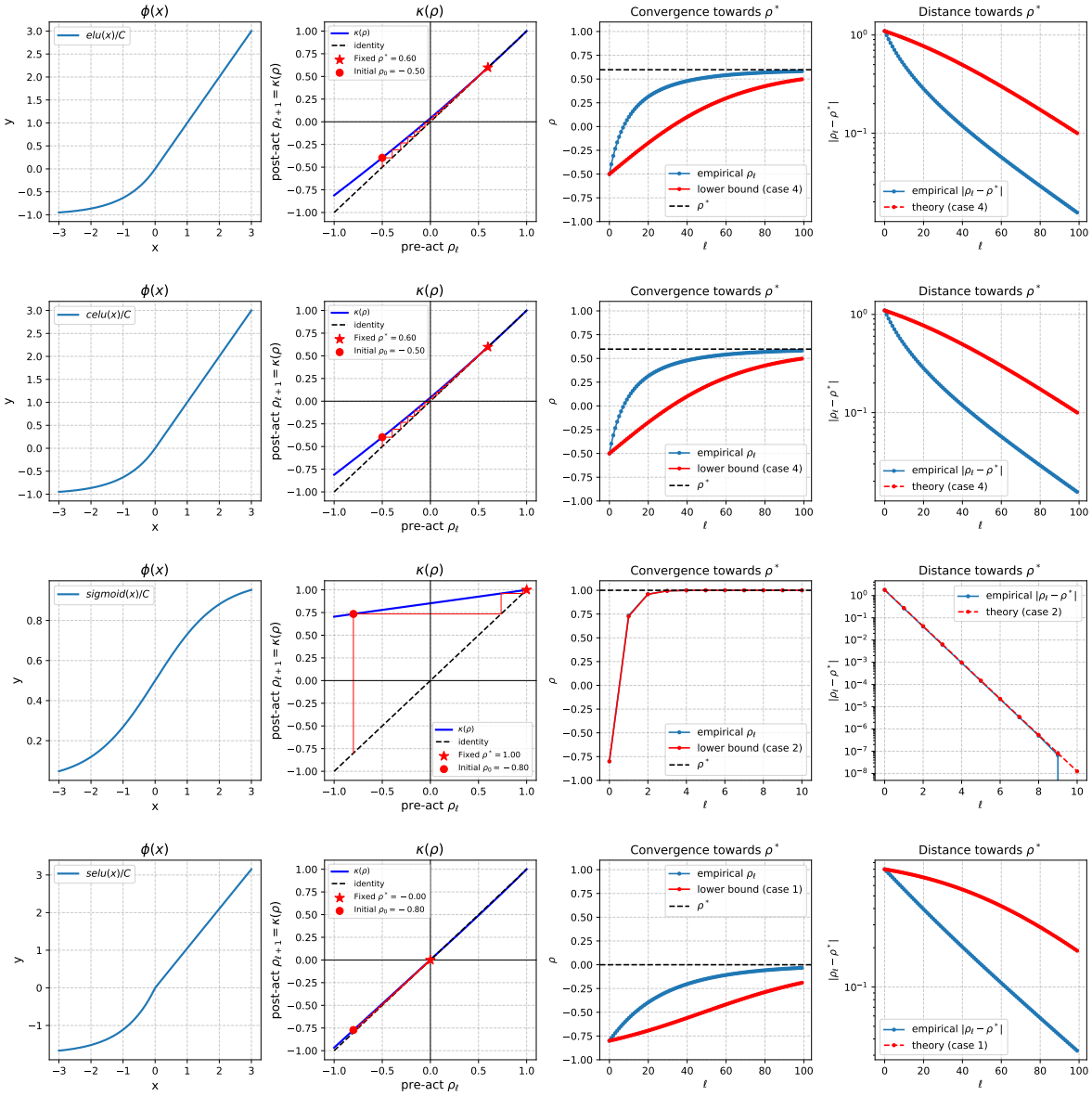


Figure S.2: Continuation of FigureS.1, for activations elu, celu, sigmoid, and selu. for the particular case of sigmoid, the errors fall below the numerical precision and cannot be computed.

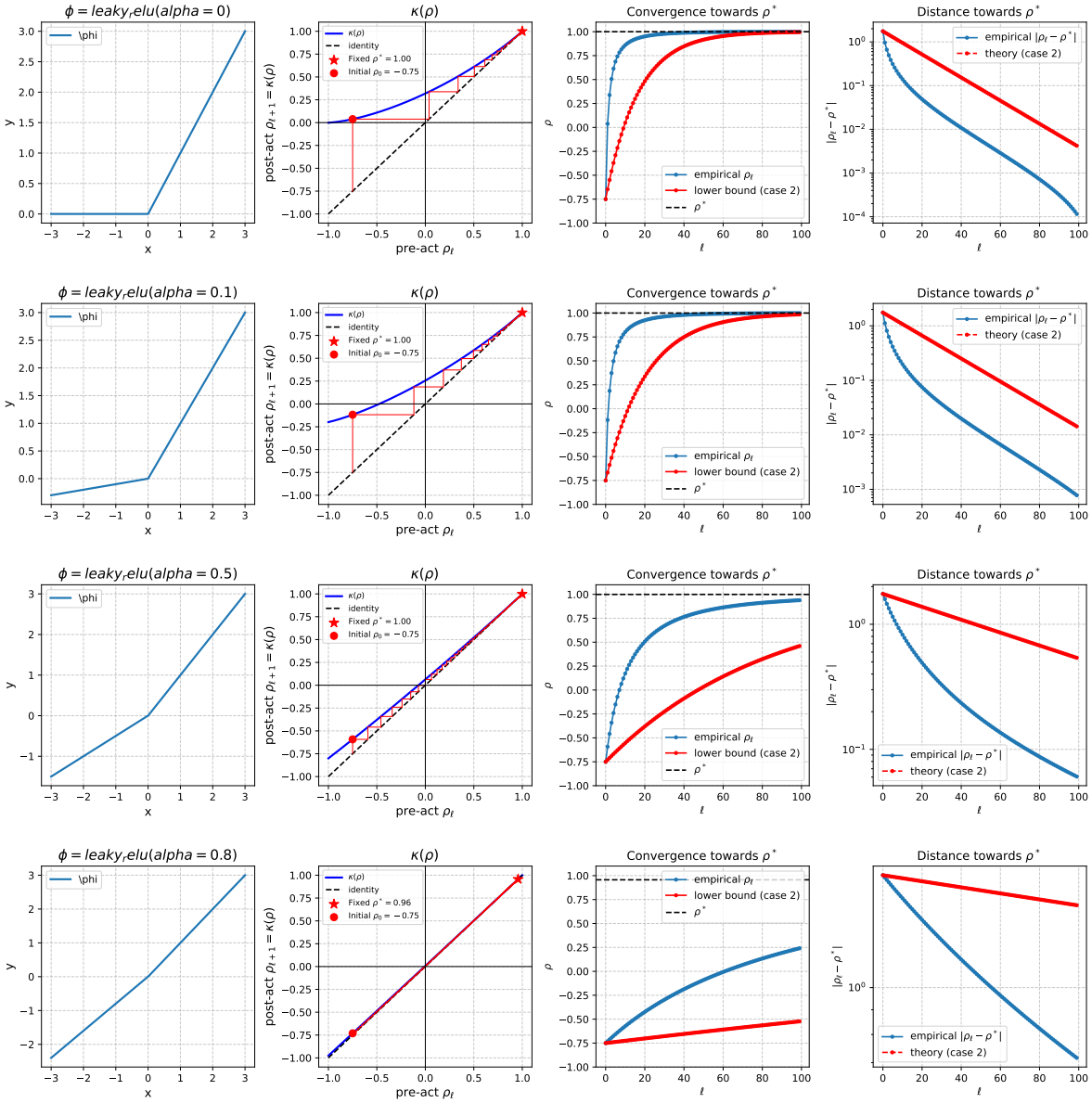


Figure S.3: Continuation of FigureS.1, for `LeakyReLU` with various negative slopes.